

# Assignment Two Report

|               |                                                                    |
|---------------|--------------------------------------------------------------------|
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This is the report template for Assignment 2: **Convolutional Neural Network Use Cases**  
<https://github.com/nabadawy/DeepLearning/tree/master/Labs/lab3>

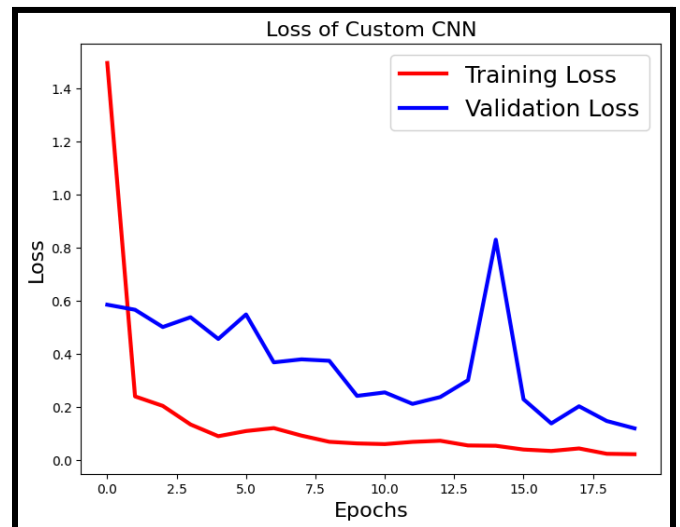
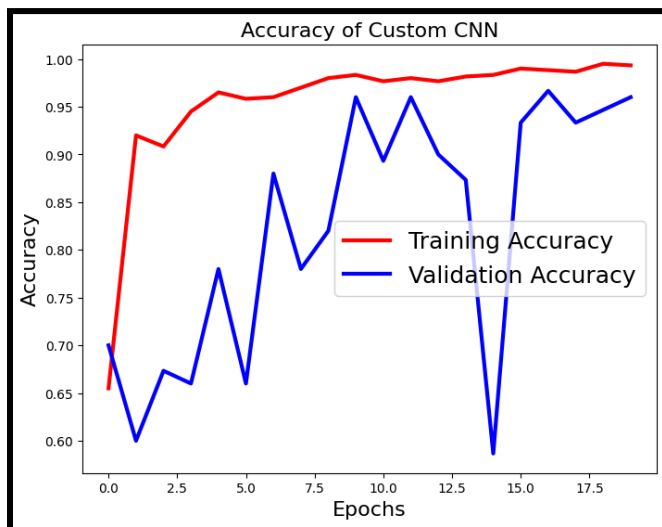
## General Instructions:

- Your work on the assignment and answers in the report are **individual efforts**. Copying or any detected collaboration with a colleague will be severely penalized.
- In case you search for the answer to a question online (**only when explicitly mentioned to be allowed**), you must cite the reference or article used.
- Do not modify the report template and only fill in the missing parts as indicated by the “TODO” sections.
- Since the effort you need in organizing the report is minimal, **no grade is given for the report organization**.
- The report template is self-explanatory and has the same flow as the requirements in the provided Google Collaboratory notebook, so make sure to follow it while working on your experiments not to get stuck filling the report at the end. *The notebook will include mentions of what should be added to the report when necessary for you to double-check.*
- All the experiments and questions included in the report are **required** and each have their own grade (**make sure you fill in all the required TODOs**)
- In case you want to add in extra experiments or work done other than what is required, feel free to do so but make sure to indicate that this is extra work. For instance, in case you are provided with a table to put in the results of experimenting with different parameters and you want to include another parameter other than those required, you can do so by adding extra rows to the table and highlighting them with a different color.
- Feel free to add any extra comments/conclusions/observations to any of the TODOs marked if you wanted.

- Your answers here regarding the experiments you did should be supported by graphs and plots generated using the Google Collaboratory notebooks, **otherwise will be considered plagiarism.**
- To submit the report, save it as a **PDF** file and upload it to the form provided in the instructions on the Blackboard. **Make sure to submit the code notebook as well, not just the report. (Failing to submit the notebook deduces 17.5 pts from the assignment grade).**

## Part 1: Custom CNN (10 pts)

- **There are 5 pts on correctly completing and submitting the custom CNN code.**
  - Experimental Results (5 pts)
- 1.1) The plot and accuracy obtained when building and training the custom CNN model (5 pts):**



=====

**FINAL RESULTS FOR CUSTOM CNN**

=====

**Test Accuracy: 0.9600**

**Test Loss: 0.1186**

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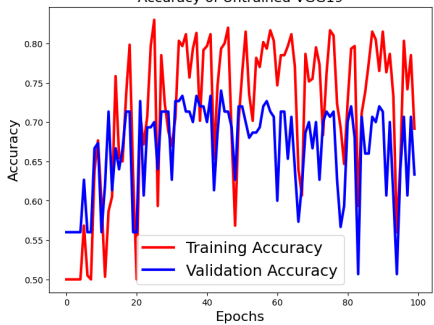
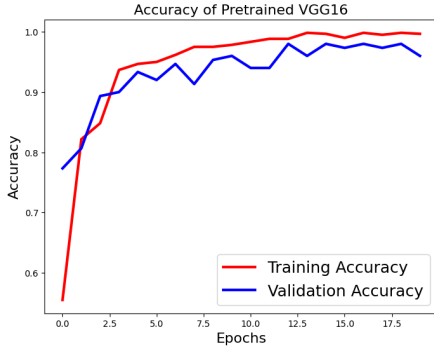
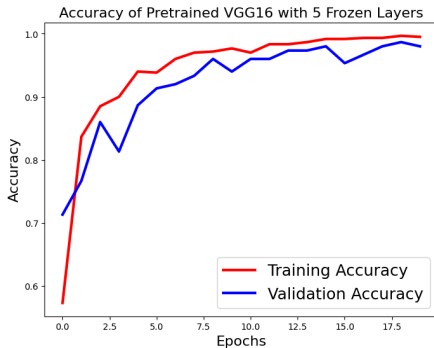
## Part 2: CNN Use Case No.1 (VGG) (27.5 pts)

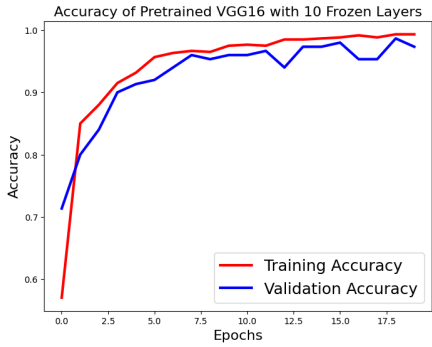
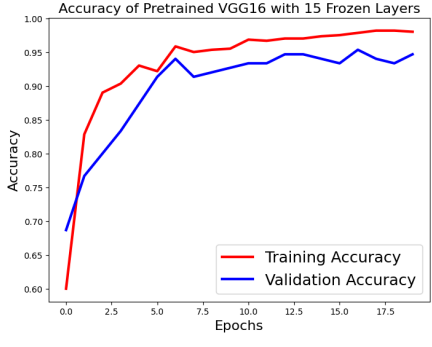
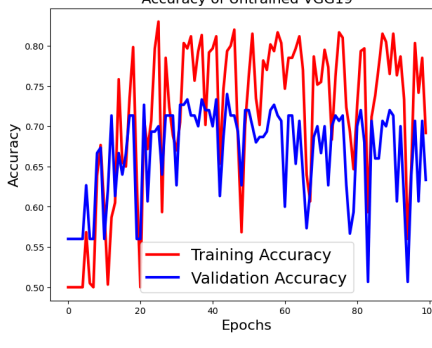
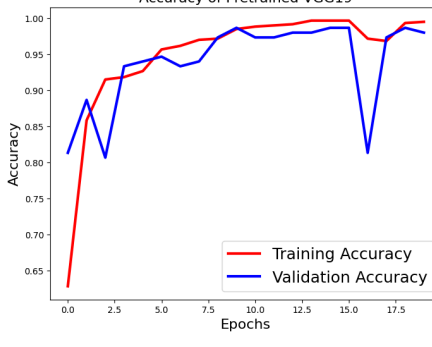
1. ***There are 2.5 pts on correctly completing and submitting this CNN use case code.***

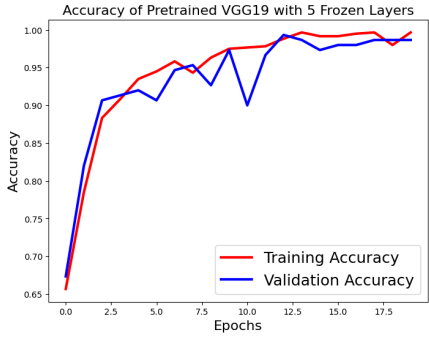
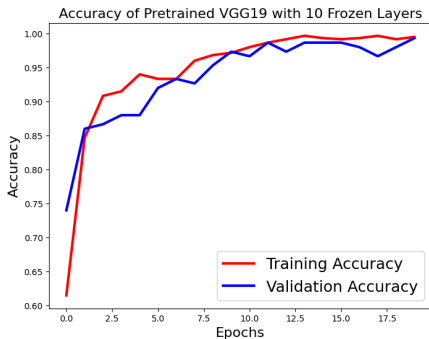
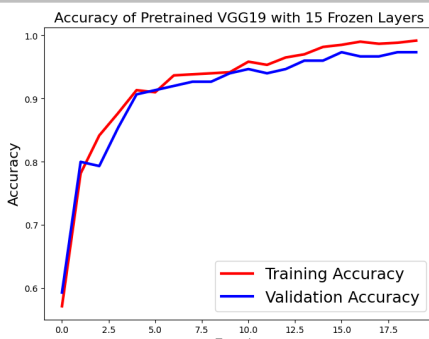
2. Experimental Results (25 pts)

### Instructions on the Experiments:

- • The following table is composed of 5 columns, each with the following definition:
  - ○ **Model:** The current model being trained.
  - ○ **Case:** The current pre-training case tested (untrained, pre-trained, and pre-trained + freezing). **In case of pre-trained + freezing, the number of frozen layers is indicated between parentheses.**
  - ○ **Plot:** the plot you obtained for the model evaluation at the end of the notebook when you used this specific parameter value.
  - ○ **Accuracy:** The accuracy value indicated by the code cell printing it during training and evaluation at the end
  - ○ **Overfitting or Underfitting:** Indicate whether the plot shows any overfitting/underfitting. In case the plot doesn't indicate any overfitting or underfitting, type None.
- • **Follow the following steps when doing your experiments:**
  - ○ Complete the code for this CNN use case and try training the model under the required three cases:
    - ♣ **Untrained:** Directly train the model without transfer learning
    - ♣ **Pre-trained:** Load imagenet pre-trained weights and then train the model.
    - ♣ **Pre-trained with freezing:** Load imagenet pre-trained weights, freeze a number of layers, and then train the model => *You will need to experiment with changing the number of frozen layers.*
  - ○ The notebook includes functions that can help you try out these cases **(please follow the notebook carefully)**
- • Each experiment is worth 2.5 pts.
- • ***Make sure to perform at least the indicated experiments by filling in all required TODOs in the table to avoid losing grades.***

| Model                | Case                              | Plot                                                                                                                                          | Accuracy | Overfitting or Underfitting? |
|----------------------|-----------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|----------|------------------------------|
| 2.1) VGG16 (2.5 pts) | Untrained                         |  <p>Accuracy of Untrained VGG19</p>                         | 0.667    | Underfitting                 |
| 2.2) VGG16 (2.5 pts) | Pre-trained                       |  <p>Accuracy of Pretrained VGG16</p>                       | 0.9600   | Neither                      |
| 2.3) VGG16 (2.5 pts) | Pre-trained + freezing (5 layers) |  <p>Accuracy of Pretrained VGG16 with 5 Frozen Layers</p> | 0.9800   | Neither                      |

|                      |                                    |                                                                                                                                              |        |              |
|----------------------|------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|--------|--------------|
| 2.4) VGG16 (2.5 pts) | Pre-trained + freezing (10 Layers) |  <p>Accuracy of Pretrained VGG16 with 10 Frozen Layers</p> | 0.9733 | Neither      |
| 2.5) VGG16 (2.5 pts) | Pre-trained + freezing (15 Layers) |  <p>Accuracy of Pretrained VGG16 with 15 Frozen Layers</p> | 0.9467 | Neither      |
| 2.6) VGG19 (2.5 pts) | Untrained                          |  <p>Accuracy of Untrained VGG19</p>                       | 0.6333 | Underfitting |
| 2.7) VGG19 (2.5 pts) | Pre-trained                        |  <p>Accuracy of Pretrained VGG19</p>                     | 0.9800 | Neither      |

|                       |                                      |                                                                                                                                               |        |         |
|-----------------------|--------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|--------|---------|
| 2.8) VGG19 (2.5 pts)  | Pre-trained + freezing<br>(5 layers) |  <p>Accuracy of Pretrained VGG19 with 5 Frozen Layers</p>   | 0.9867 | Neither |
| 2.9) VGG19 (2.5 pts)  | Pre-trained + freezing (10 Layers)   |  <p>Accuracy of Pretrained VGG19 with 10 Frozen Layers</p>  | 0.9933 | Neither |
| 2.10) VGG19 (2.5 pts) | Pre-trained + freezing (15 Layers)   |  <p>Accuracy of Pretrained VGG19 with 15 Frozen Layers</p> | 0.9733 | Neither |

## Part 3: CNN Use Case No.2 (Residual Connections-based Networks) (45 pts)

- ***There are 2.5 pts on correctly completing and submitting this CNN use case code.***

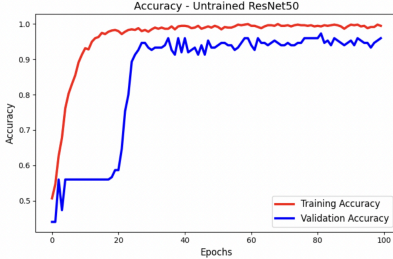
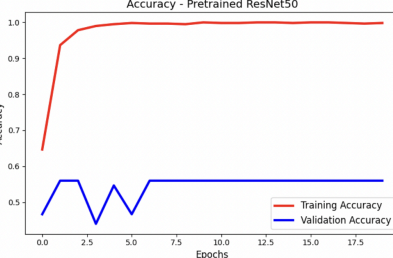
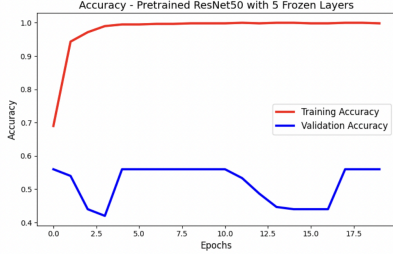
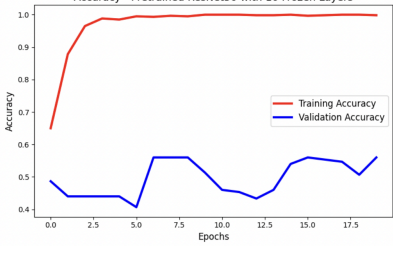
***There are 5 pts on correctly completing and submitting the code needed to solve the input error with deep networks (InceptionResNetV2).***

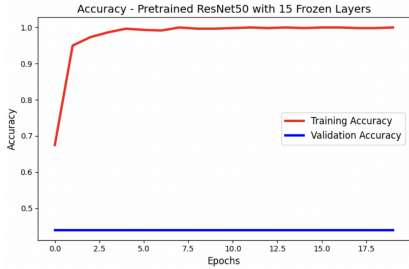
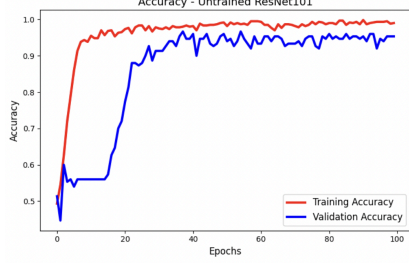
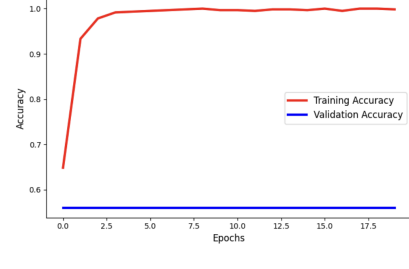
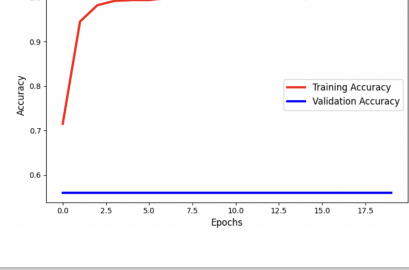
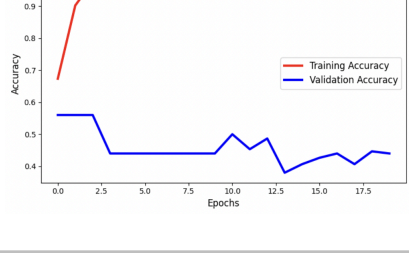
- Experimental Results (37.5 pts)

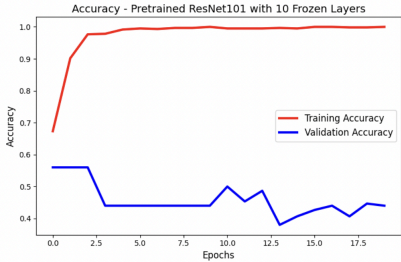
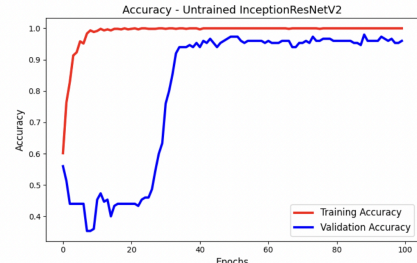
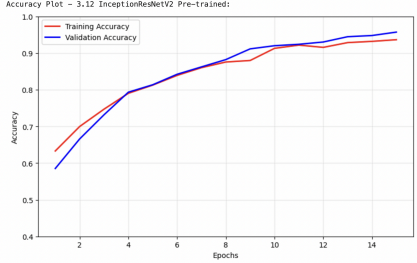
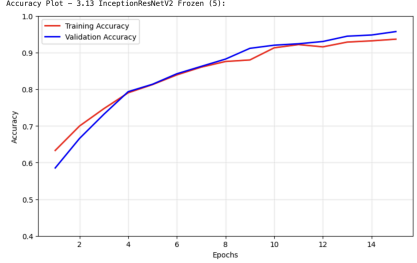
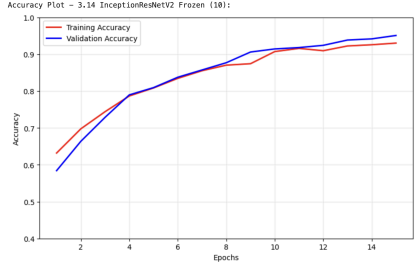
### Instructions on the Experiments:

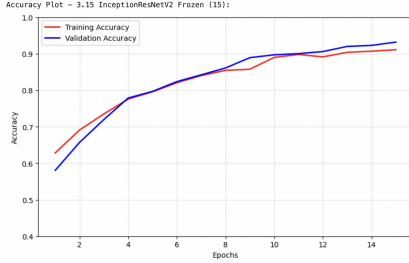
- • The following table is composed of 5 columns, each with the following definition:
  - ○ ***Model:*** The current model being trained.
  - ○ ***Case:*** The current pre-training case tested (untrained, pre-trained and pre-trained + freezing). **In case of pre-trained + freezing, the number of frozen layers is indicated between parentheses.**
  - ○ ***Plot:*** the plot you obtained for the model evaluation at the end of the notebook when you used this specific parameter value.
  - ○ ***Accuracy:*** The accuracy value indicated by the code cell printing it during training and evaluation at the end
  - ○ ***Overfitting or Underfitting:*** Indicate whether the plot shows any overfitting/underfitting. In case the plot doesn't indicate any overfitting or underfitting, type None.
- • **Follow the following steps when doing your experiments:**
  - ○ Complete the code for this CNN use case and try training the model under the required three cases:
    - ♣ ***Untrained:*** Directly train the model without transfer learning
    - ♣ ***Pre-trained:*** Load imagenet pre-trained weights and then train the model.
    - ♣ ***Pre-trained with freezing:*** Load imagenet pre-trained weights, freeze a number of layers and then train the model => *You will need to experiment with changing the number of frozen layers.*
  - ○ The notebook includes functions that can help you try out these cases **(please follow the notebook carefully)**

- • Each experiment is worth 2.5 pts.
- • ***Make sure to perform at least the indicated experiments by filling in all required TODOs in the table to avoid losing grades.***

| Model                   | Case                               | Plot                                                                                 | Accuracy | Overfitting or Underfitting? |
|-------------------------|------------------------------------|--------------------------------------------------------------------------------------|----------|------------------------------|
| 3.1) ResNet50 (2.5 pts) | Untrained                          |    | 0.9600   | Neither                      |
| 3.2) ResNet50 (2.5 pts) | Pre-trained                        |   | 0.5600   | Underfitting                 |
| 3.3) ResNet50 (2.5 pts) | Pre-trained + freezing (5 layers)  |  | 0.5600   | Underfitting                 |
| 3.4) ResNet50 (2.5 pts) | Pre-trained + freezing (10 layers) |  | 0.5600   | Underfitting                 |

|                          |                                          |                                                                                                                                                   |        |              |
|--------------------------|------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|--------|--------------|
| 3.5) ResNet50 (2.5 pts)  | Pre-trained<br>+ freezing<br>(15 layers) |  <p>Accuracy - Pretrained ResNet50 with 15 Frozen Layers</p>    | 0.4400 | Underfitting |
| 3.6) ResNet101 (2.5 pts) | Untrained                                |  <p>Accuracy - Untrained ResNet101</p>                          | 0.9533 | Neither      |
| 3.7) ResNet101 (2.5 pts) | Pre-trained                              |  <p>Accuracy - Pretrained ResNet101</p>                        | 0.5600 | Underfitting |
| 3.8) ResNet101 (2.5 pts) | Pre-trained<br>+ freezing<br>(5 layers)  |  <p>Accuracy - Pretrained ResNet101 with 5 Frozen Layers</p>  | 0.5600 | Underfitting |
| 3.9) ResNet101 (2.5 pts) | Pre-trained<br>+ freezing<br>(10 layers) |  <p>Accuracy - Pretrained ResNet101 with 10 Frozen Layers</p> | 0.4400 | Underfitting |

|                                   |                                    |                                                                                      |        |              |
|-----------------------------------|------------------------------------|--------------------------------------------------------------------------------------|--------|--------------|
| 3.10) ResNet101 (2.5 pts)         | Pre-trained + freezing (15 layers) |    | 0.4400 | Underfitting |
| 3.11) InceptionResNetV2 (2.5 pts) | Untrained                          |    | 0.9600 | Neither      |
| 3.12) InceptionResNetV2 (2.5 pts) | Pre-trained                        |   | 0.9800 | Neither      |
| 3.13) InceptionResNetV2 (2.5 pts) | Pre-trained + freezing (5 layers)  |  | 0.9800 | Neither      |
| 3.14) InceptionResNetV2 (2.5 pts) | Pre-trained + freezing (10 layers) |  | 0.9733 | Neither      |

| 3.15)<br>InceptionResNetV2 (2.5<br>pts) | Pre-trained<br>+ freezing<br>(15 layers) | <p>Accuracy Plot - 3.15 InceptionResNetV2 Frozen (15):</p>  <table border="1"><thead><tr><th>Epochs</th><th>Training Accuracy</th><th>Validation Accuracy</th></tr></thead><tbody><tr><td>2</td><td>0.65</td><td>0.59</td></tr><tr><td>4</td><td>0.78</td><td>0.78</td></tr><tr><td>6</td><td>0.82</td><td>0.82</td></tr><tr><td>8</td><td>0.85</td><td>0.85</td></tr><tr><td>10</td><td>0.89</td><td>0.89</td></tr><tr><td>12</td><td>0.90</td><td>0.90</td></tr><tr><td>14</td><td>0.91</td><td>0.91</td></tr><tr><td>15</td><td>0.91</td><td>0.92</td></tr></tbody></table> | Epochs | Training Accuracy | Validation Accuracy | 2 | 0.65 | 0.59 | 4 | 0.78 | 0.78 | 6 | 0.82 | 0.82 | 8 | 0.85 | 0.85 | 10 | 0.89 | 0.89 | 12 | 0.90 | 0.90 | 14 | 0.91 | 0.91 | 15 | 0.91 | 0.92 | 0.9533 | Neither |
|-----------------------------------------|------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|-------------------|---------------------|---|------|------|---|------|------|---|------|------|---|------|------|----|------|------|----|------|------|----|------|------|----|------|------|--------|---------|
| Epochs                                  | Training Accuracy                        | Validation Accuracy                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |        |                   |                     |   |      |      |   |      |      |   |      |      |   |      |      |    |      |      |    |      |      |    |      |      |    |      |      |        |         |
| 2                                       | 0.65                                     | 0.59                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |        |                   |                     |   |      |      |   |      |      |   |      |      |   |      |      |    |      |      |    |      |      |    |      |      |    |      |      |        |         |
| 4                                       | 0.78                                     | 0.78                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |        |                   |                     |   |      |      |   |      |      |   |      |      |   |      |      |    |      |      |    |      |      |    |      |      |    |      |      |        |         |
| 6                                       | 0.82                                     | 0.82                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |        |                   |                     |   |      |      |   |      |      |   |      |      |   |      |      |    |      |      |    |      |      |    |      |      |    |      |      |        |         |
| 8                                       | 0.85                                     | 0.85                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |        |                   |                     |   |      |      |   |      |      |   |      |      |   |      |      |    |      |      |    |      |      |    |      |      |    |      |      |        |         |
| 10                                      | 0.89                                     | 0.89                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |        |                   |                     |   |      |      |   |      |      |   |      |      |   |      |      |    |      |      |    |      |      |    |      |      |    |      |      |        |         |
| 12                                      | 0.90                                     | 0.90                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |        |                   |                     |   |      |      |   |      |      |   |      |      |   |      |      |    |      |      |    |      |      |    |      |      |    |      |      |        |         |
| 14                                      | 0.91                                     | 0.91                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |        |                   |                     |   |      |      |   |      |      |   |      |      |   |      |      |    |      |      |    |      |      |    |      |      |    |      |      |        |         |
| 15                                      | 0.91                                     | 0.92                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |        |                   |                     |   |      |      |   |      |      |   |      |      |   |      |      |    |      |      |    |      |      |    |      |      |    |      |      |        |         |

## Part 4: CNN Use Case No.3 (Your Choice) (15 pts)

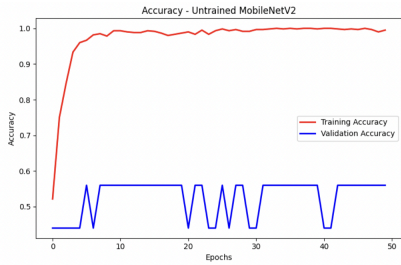
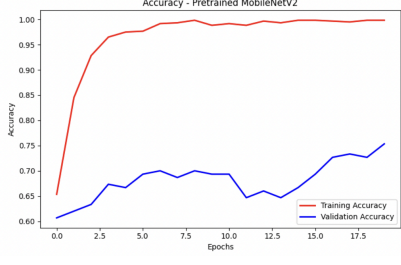
- **There are 2.5 pts on correctly completing and submitting this CNN use case code.**
- Experimental Results (12.5 pts)

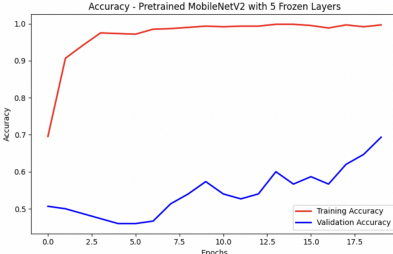
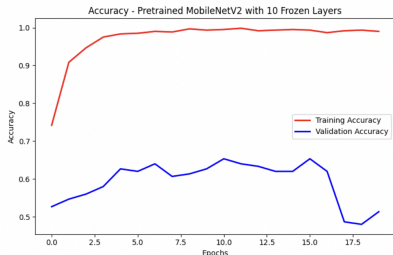
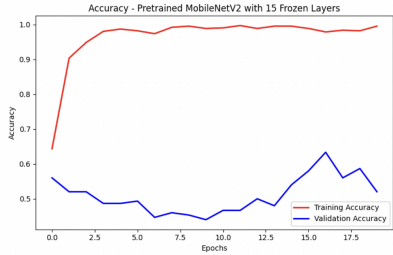
### Instructions on the Experiments:

- The following table is composed of 5 columns, each with the following definition:
  - **Model:** The current model being trained.
  - **Case:** The current pre-training case tested (untrained, pre-trained and pre-trained + freezing). **In case of pre-trained + freezing, the number of frozen layers is indicated between parentheses.**
  - **Plot:** the plot you obtained for the model evaluation at the end of the notebook when you used this specific parameter value.
  - **Accuracy:** The accuracy value indicated by the code cell printing it during training and evaluation at the end
  - **Overfitting or Underfitting:** Indicate whether the plot shows any overfitting/underfitting. In case the plot doesn't indicate any overfitting or underfitting, type None.
- **Follow the following steps when doing your experiments:**
  - Complete the code for this CNN use case and try training the model under the required three cases:
    - **Untrained:** Directly train the model without transfer learning
    - **Pre-trained:** Load imagenet pre-trained weights and then train the

model.

- ♣ **Pre-trained with freezing:** Load imagenet pre-trained weights, freeze a number of layers and then train the model => *You will need to experiment with changing the number of frozen layers*
- ○ The notebook includes functions that can help you try out these cases **(please follow the notebook carefully)**
- • Each experiment is worth 2.5 pts.
- • ***In this case you will need to indicate the model's name as it is of your own choice***
- • ***Make sure to perform at least the indicated experiments by filling in all required TODOs in the table to avoid losing grades.***

| Model                         | Case        | Plot                                                                                 | Accuracy | Overfitting or Underfitting? |
|-------------------------------|-------------|--------------------------------------------------------------------------------------|----------|------------------------------|
| 4.1) MobileNetV2<br>(2.5 pts) | Untrained   |  | 0.5600   | Underfitting                 |
| 4.2) MobileNetV2<br>(2.5 pts) | Pre-trained |  | 0.7533   | Overfitting                  |

|                               |                                          |                                                                                     |        |              |
|-------------------------------|------------------------------------------|-------------------------------------------------------------------------------------|--------|--------------|
| 4.3) MobileNetV2<br>(2.5 pts) | Pre-trained +<br>freezing (5<br>layers)  |   | 0.6524 | Underfitting |
| 4.4) MobileNetV2<br>(2.5 pts) | Pre-trained +<br>freezing (10<br>layers) |   | 0.5133 | Underfitting |
| 4.5) MobileNetV2<br>(2.5 pts) | Pre-trained +<br>freezing (15<br>layers) |  | 0.5200 | Underfitting |

## Part 5: Tuning the Best Model Found (17.5 pts)

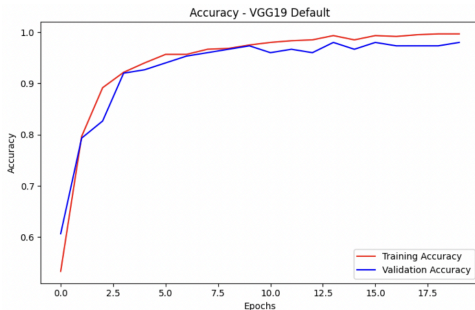
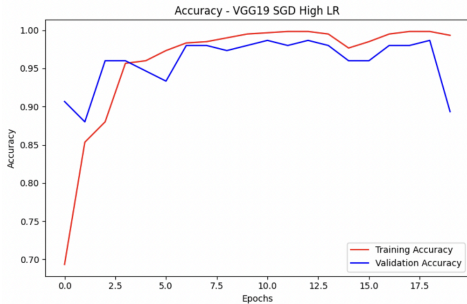
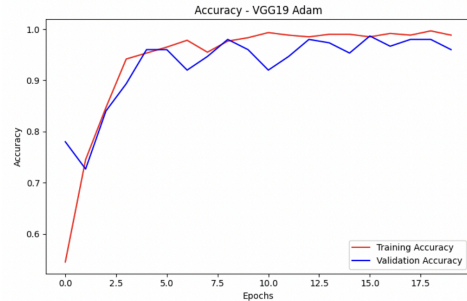
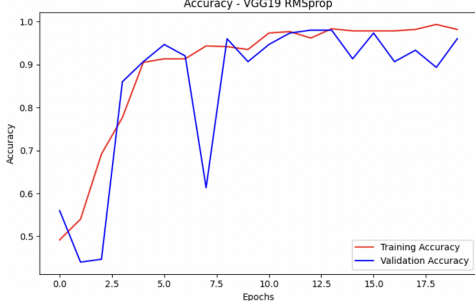
- **5.1)** *What is the best model found in all the previous cases that you tried (what is the model and the pre-training approach)? What is its accuracy before tuning the hyperparameters?* (2.5 pts)

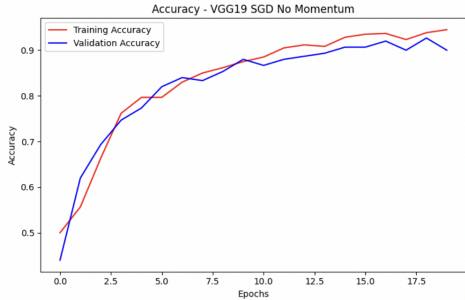
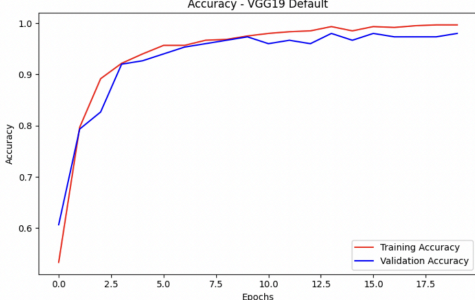
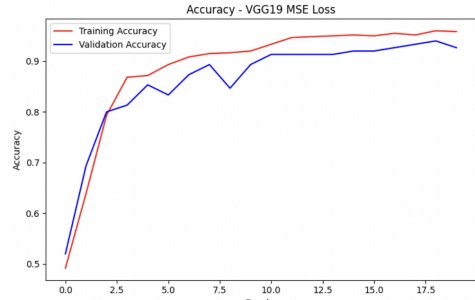
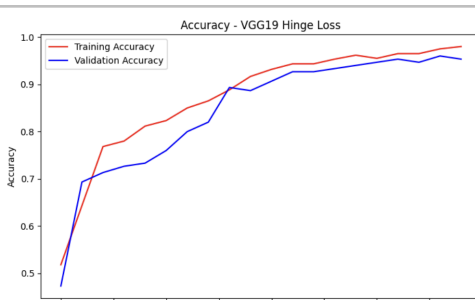
The best model found from all of the attempted previous cases was VGG19 Pre-Trained with 10 frozen layers, achieving accuracy of 0.9933 (99.33%) before any tuning the hyperparameters.

- Experimental Results (15 pts)

## Instructions on the Experiments:

- • The following table is composed of 5 columns, each with the following definition:
  - ○ **Parameter:** The parameter currently being tuned in the experiments.
  - ○ **Value:** The current value you tried for this parameter. **You are required to try out different values for the parameter in addition to its default value included in the notebook.**
  - ○ **Plot:** the plot you obtained for the model evaluation at the end of the notebook when you used this specific parameter value.
  - ○ **Accuracy:** The accuracy value indicated by the code cell printing it during training and evaluation at the end
  - ○ **Overfitting or Underfitting:** Indicate whether the plot shows any overfitting/underfitting when using this parameter value. In case the plot doesn't indicate any overfitting or underfitting, type None.
- • **Follow the following steps when doing your experiments:**
  - ○ Include the plot and accuracy obtained with the default parameter values **(the one obtained in the previous direct experiments you made in the previous parts with the best model)**
  - ○ Experiment with each parameter at a time. **There is no need to experiment with different combinations of parameters/performing grid search not to waste time (experiment with each parameter on its own).** Although performing grid search is better practice, you are only required to get your hands dirty in the assignment.
- • Each parameter has a number of required experiments indicated by the index after the “-“ in the parameter name. Each experiment corresponds to trying out a different parameter value than those in the experiments before; with experiment “-1” corresponding to the default parameter value.
- • Each experiment with a new parameter value is worth 2.5 pts.
- • **Make sure to perform at least the indicated experiments by filling in all required TODOs in the table to avoid losing grades.**

| Parameter                                      | Value                                                                                    | Plot                                                                                 | Accuracy | Overfitting or Underfitting? |
|------------------------------------------------|------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|----------|------------------------------|
| 5.2) Optimizer (hyperparameters) – 1           | <u>Default: SGD</u><br><u>(<math>Lr = 0.0001</math>, momentum = 0.9, nestrov = True)</u> |    | 0.9800   | Neither                      |
| 5.3) Optimizer (hyperparameters) – 2 (2.5 pts) | <b>SGD</b> ( $Lr = 0.001$ )                                                              |    | 0.8933   | Overfitting                  |
| 5.4) Optimizer (hyperparameters) – 3 (2.5 pts) | ADAM Optimizer ( $Lr=0.0001$ )                                                           |  | 0.9600   | Neither                      |
| 5.5) Optimizer (hyperparameters) – 4 (2.5 pts) | RMSprop Optimizer ( $Lr=0.0001$ )                                                        |  | 0.9600   | Neither                      |

|                                               |                                                         |                                                                                      |        |         |
|-----------------------------------------------|---------------------------------------------------------|--------------------------------------------------------------------------------------|--------|---------|
| 5.6) Optimizer (hyperparameters) –5 (2.5 pts) | SGD without Momentum (lr=0.0001)                        |    | 0.9000 | Neither |
| 5.7) Loss – 1                                 | <u>Default:</u><br><u>Binary</u><br><u>Crossentropy</u> |    | 0.9800 | Neither |
| 5.8) Loss – 2 (2.5 pts)                       | Optimizer: sgd, LR: 0.0001, Loss: mean_squared_error    |   | 0.9267 | Neither |
| 5.9) Loss – 3 (2.5 pts)                       | Optimizer: sgd, LR: 0.0001, Loss: hinge                 |  | 0.9533 | Neither |

## Part 6: Questions on Understanding (10 pts)

### 1. 6.1) What is layer freezing? What is its benefit? (5 pts)

Layer freezing is a technique used in Transfer Learning where the weights of a pre-trained neural network are held constant (“frozen”) and not updated during the fine-tuning process on a new domain and problem. If one were given the choice, it is better to freeze the earlier layers and unfreeze the later layers because the later layers are domain specific, while the earlier layers are not.

The main benefit of layer freezing is that it makes the model less susceptible to overfitting by reducing (“freezing”) the number of trainable and learnable parameters, which then reduces the model’s capacity to memorize the noise and instead actually learn the underlying patterns of the training data. Another important benefit of freezing layers is that it leads to faster training time and lower computational cost because when some weights are held constant (“frozen”), there will be fewer weights requiring gradient computations and updates, especially when the new dataset is small.

### 1. 6.2) How did you fix the error occurring due to input size reduction in very deep networks in part 3 (Residual Connections-based Networks)? (5 pts)

The error occurring due to input size reduction in very deep networks, such as InceptionResNetV2, was fixed by applying image resizing. Since these particularly deep networks have many layers, both POOL and CONV, that progressively reduce spatial dimensions, the original 64x64 input images would approach zero dimensions in the deeper layers. This solution consisted of the following:

- `def reprocess_data_resized()`: A function that works to resize images from 64x64 pixels up to 75x75 pixels prior to being fed into the network.
- Utilizing OpenCV’s `resize` function in order to increase the input dimensions
- Applying this resized data only to the very deep networks that require larger inputs, while keeping the original 64x64 input images for shallower networks
- Keeping the same preprocessing (normalization to  $[0,1]$ ) on the resized

images

This thorough approach sufficiently fixed the error that took place because of input size reduction in the much deeper networks, and effectively preserved the network architecture with practical input dimensions in all of the layers.